

A
Problem Based Learning-Report On

IOT BASED SMART PLUG ENERGY METER

In partial fulfillment of requirements for the award of

**BACHELOR OF TECHNOLOGY
IN
ELECTRICAL AND ELECTRONICS ENGINEERING**

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2025–2026

HYDERABAD INSTITUTE OF TECHNOLOGY AND MANAGEMENT
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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING



CERTIFICATE

This is to certify that the project work entitled “**IOT BASED SMART PLUG ENERGY METER**” is a Bonafide work carried out by **CH.SANDHYA** bearing Roll No. **24E51A0209**, **K.HARIKA** bearing Roll No.**24E51A0216**, **KUDIPUDI PAVANI** bearing Roll No.**24E51A0218**, **G.TEJASWI** bearing the Roll no. **25E55A0201** ,bearing in partial fulfilment of the requirements for the degree BACHELOR OF TECHNOLOGY in EEE by the Jawaharlal Nehru Technological University, Hyderabad, during the academic year 2025-2026. The matter contained in this document has not been submitted to any other University or institute for the award of any degree or diploma.

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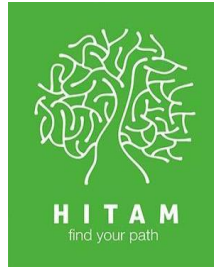
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DECLARATION

We CH.SANDHYA (24E51A0209), K.HARIKA (24E51A0216), KUDIPUDI PAVANI (24E51A0218), G.TEJASWI (25E55A0201), students of Bachelor of Technology in EEE, session: 2025-2026, Hyderabad Institute of Technology and Management, Dundigal, Hyderabad, Telangana State. Hereby declare that the work presented in this project entitled “**IOT BASED SMART PLUG ENERGY METER**” is the outcome of our own Bonafide work and is correct to the best of our knowledge. This work has been undertaken with adherence to engineering ethics. It contains no material previously published or written by another person, nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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ACKNOWLEDGEMENT

An endeavor of a long period can be successful only with the advice of many well-wishers.

We would like to thank our chairman **Sri ARUTLA PRASHANTH** for providing all the facilities to carry out project work successfully.

We would like to thank our Principal **Dr. S. ARAVIND** who has inspired a lot through their speeches and providing this opportunity to carry out our Mini project successfully.

We are very thankful to our Head of the Department, Dr O.P SURESH

We would like to specially thank our **Mr.S.V.SATYANARAYANA** Dept of EEE, Associate Professor and coordinator of the Problem Based Learning {**PBL**} who is constant encouragement and enormous support provided to us for carrying out PBL project work.

We wish to convey our gratitude and express sincere thanks to all **D.C SSDC and P.R.C (Project Review Committee)** members, non teaching staff for their support and Cooperation rendered for successful submission of our project work.

We also want to express our sincere gratitude to all my family members and my friends for their individual care and everlasting moral support.

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PROBLEM

In many places like schools, offices, shops, and even small businesses, fans, lights, and appliances are often left on even when no one is using them. This leads to a lot of wasted electricity and higher bills, simply because people don't have an easy way to track what's being used in real time. Most energy monitoring devices available today are either too expensive or too complicated for everyday use. What's really needed is a simple, low-cost system that can show how much energy is being used and give quick alerts—like a reminder to switch things off—through basic circuits. This would make it easier for people to save power, cut down costs, and avoid wasting electricity without needing any technical knowledge.

CHAPTER-1

INTRODUCTION

This project focuses on developing an **IoT-based Smart Plug Energy Meter**, a compact device that monitors the real-time electricity consumption of any connected household appliance. The system tracks voltage, current, power, and total energy usage, giving users clear visibility into how much electricity each device consumes. Instead of relying on costly commercial energy meters, this model offers a **low-cost, scalable, and practical alternative** for homes, hostels, and small offices.

The smart plug integrates **voltage and current sensing modules** with an **ESP32 microcontroller**, which processes analog readings and computes real-time power data. This information is then pushed to the cloud and displayed on a mobile dashboard through IoT platforms such as Blynk or MQTT. Users can remotely monitor consumption, control the appliance through the plug's internal relay, and receive alerts when abnormal or excessive energy usage is detected.

Key features include:

- Accurate real-time measurement of voltage, current, power, and energy.
- Remote ON/OFF control of appliances via mobile app.
- Alerts for overload, abnormal consumption, or faulty appliance behavior.
- Data logging for analyzing energy patterns over time.
- Low-cost design using easily available sensors and an ESP32 module.

This project helps users understand device-level power behavior and encourages energy-efficient habits. It also strengthens practical knowledge of IoT systems, electrical measurements, embedded programming, and smart-home automation. By offering an affordable, intelligent, and user-friendly solution, the system supports smarter energy management in modern households.

CHAPTER 2

EXISTING SOLUTION

Existing solutions for monitoring appliance-level electricity usage mainly involve commercial **IoT smart plugs** and advanced home-automation energy meters. These devices provide accurate real-time consumption data, app-based monitoring, and automated alerts. However, they are **high-cost, proprietary, and often over-engineered**, making them inaccessible for students, budget-restricted users, and households seeking an affordable alternative.

Most households still depend on **standard digital energy meters**, which show only total kWh usage. They do not indicate which appliance consumes how much power, nor do they provide real-time load variations, overload alerts, or remote control options. As a result, users only notice excessive consumption after receiving the monthly electricity bill.

Commercial smart plugs and IoT home-energy systems do offer appliance-level monitoring, but they rely on complex internally-coded firmware, cloud integrations, premium dashboards, and paid ecosystems. Their cost increases due to Wi-Fi modules, proprietary control circuits, and certified sensing hardware, making them unsuitable for low-cost educational or prototype-level deployment.

In contrast, the proposed **IoT-based Smart Plug Energy Meter** provides the same core functionality using **low-cost sensors, ESP32, basic measurement circuits, and an open-source dashboard**. It delivers real-time voltage, current, power, and energy readings while keeping the design affordable, simple to build, and easy to install. This makes it practical for households, hostels, and small communities that need reliable energy monitoring without the complexity and high price of commercial solutions.

LITERATURE SURVEY

1) **Design and Implementation of IoT-based Energy Efficient Smart Metering System for Domestic Applications**

Traditional energy meters only measure cumulative consumption and lack real-time or appliance-level monitoring. IoT-based smart meters using ESP32/ESP8266, ACS712 current sensor, and ZMPT101B voltage sensor enable real-time monitoring, alerts, and optional cloud connectivity. Appliance-level tracking helps identify high-energy devices and promotes efficient usage.

2) **Design and Implementation of an Internet of Things (IoT) Based Smart Energy Metering**

Energy monitoring is crucial to reduce electricity wastage and manage consumption efficiently. Traditional analog and digital meters only provide cumulative usage and lack real-time insights. IoT-based smart meters, using microcontrollers like **ESP32/ESP8266** with **ACS712 current sensors** and **ZMPT101B voltage sensors**, allow **real-time monitoring, logging, and alerts**.

CHAPTER 3

PROPOSED SOLUTION

The proposed solution is an IoT-based Smart Plug Energy Meter that provides low-cost, real-time monitoring of appliance power consumption. It uses voltage and current sensors with an ESP32 to measure voltage, current, power, and energy. Data is sent to a mobile dashboard, where users can track usage, receive overload or abnormal-consumption alerts, and remotely switch the appliance ON/OFF through the plug's relay. The system delivers accurate monitoring, remote control, and energy-saving insights while staying affordable and easy to install.

Advanced Features

1. Real-Time Data Logging and Cloud Storage

The smart plug records voltage, current, power, and energy using the ESP32. Data can be stored locally (EEPROM) or sent to cloud dashboards for long-term tracking and analysis.

2. IoT-Based Wireless Monitoring

Through Wi-Fi connectivity, the plug sends live consumption data to a mobile app. Users can view graphs, receive alerts, and monitor the appliance remotely from anywhere.

3. Appliance-Level Energy Tracking

Since each smart plug is attached to a single device, it provides precise appliance-level consumption. Users can identify high-load appliances and receive device-specific alerts.

4. Predictive Alerts and Smart Insights

The system analyzes usage patterns to estimate daily or monthly consumption. It warns users when they approach preset limits and provides suggestions to reduce unnecessary energy usage.

5. Automatic Load Protection and Control

The plug can automatically switch OFF the appliance when overload, abnormal current, or unsafe voltage levels are detected. This prevents damage and avoids excessive energy consumption.

Prototype Development

The prototype for the IoT-based Smart Plug Energy Meter was developed in a structured manner, starting with sensing hardware design and ending with IoT integration and relay-based control. The objective of this prototype is to measure appliance-level voltage, current, power, and energy, and to provide real-time monitoring and protection through a mobile dashboard.

Sensor Integration and Measurement Circuit Design

A. Voltage Measurement

A stepped-down AC voltage sampling circuit is connected to the plug's input line. The voltage is scaled to a safe range and fed into the ESP32 ADC. Isolation is maintained through transformer or optocoupler-based safety stages.

B. Current Measurement

An ACS712 or SCT-013 current sensor is placed in series with the appliance load. The sensor outputs an analog voltage corresponding to the real-time load current. This provides accurate and isolated current measurement for the smart plug.

C. Power & Energy Computation

The ESP32 reads voltage and current through ADC pins. Using these values, it calculates:

$$P = V \times I$$

Energy is computed by integrating power over time to determine Wh/kWh consumption.

2. ESP32 Programming and IoT Logic

The ESP32 firmware performs the following operations:

- Continuous ADC sampling for voltage and current
- Noise filtering using averaging
- Real-time power and energy computation
- Overload and abnormal load detection
- Relay ON/OFF control for appliance switching
- Wi-Fi connection to send data to Blynk or cloud dashboard
- Display of live readings through mobile app graphs, gauges, and logs

This creates a complete IoT-based monitoring and control platform.

3. Hardware Assembly of the Smart Plug Prototype

Components are mounted inside a compact smart-plug enclosure:

- ESP32 fixed centrally for sensor and relay wiring
- ACS712/CT mounted near the load line
- Relay module connected to switch the output socket
- Voltage sampling circuit placed safely with insulation

- Indicator LEDs for Wi-Fi status, power status, and overload
- 5V regulator to power ESP32 and sensing modules
- Proper separation between AC high-voltage and low-voltage sections

The internal layout ensures stable operation and safety.

4. IoT Alerts and Control System

The smart plug includes intelligent alert mechanisms:

- Mobile push notifications for overload
- App-based warnings for abnormal current or voltage
- Automatic relay cutoff during unsafe load levels
- Real-time graph view of V/I/P/E parameters
- Manual and automatic ON/OFF control from the app

These alerts ensure protection and efficient energy usage.

5. Enclosure and Safety Features

The prototype is enclosed with:

- Heat-resistant plastic casing
- Ventilation slots for airflow
- Fused input line for protection
- Clear labels for input/output
- Complete AC–DC isolation
- Shock-proof internal wiring

This ensures safe handling during operation.

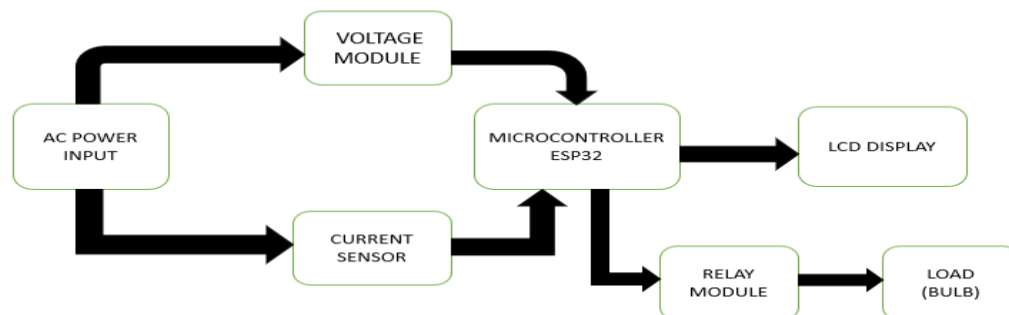
6. Testing and Calibration

The prototype is tested under different appliance loads:

- Voltage calibration using a multimeter
- Current calibration using resistive loads
- Power validation using appliances with known wattage
- Relay response testing under overload
- Wi-Fi stability and dashboard data consistency
- Long-duration tests to confirm accurate energy tracking

After calibration, the smart plug reliably measures, displays, and controls appliance-level electricity usage with stable IoT connectivity.

BLOCK DIAGRAM



WORKING

The working of the IoT-Based Smart Plug Energy Meter begins when the 230V AC supply is connected to the smart plug and an appliance is plugged into its output socket. When the appliance is switched ON, current flows through the smart plug, and the sensing stage starts operating. A current sensor such as ACS712 or SCT-013 is placed in series with the appliance load to measure the real-time current drawn. At the same time, a voltage sensing circuit samples the AC voltage from the plug input.

Both voltage and current signals are forwarded to the ESP32 microcontroller through its ADC pins. The ESP32 then processes the sampled data to compute live electrical parameters. Instantaneous power is calculated using:

$$\text{Power} = \text{Voltage} \times \text{Current}$$

The energy consumption is computed over time using:

$$\text{Energy (Wh / kWh)} = \text{Power} \times \text{Time}$$

These processed values are sent to the IoT dashboard (Blynk or MQTT) for real-time display. The microcontroller also activates protections like overload alerts and automatic relay cutoff whenever unsafe current or voltage levels are detected. This completes the monitoring and control functionality of the smart plug energy meter.

ADVANTAGES

1. **Low-Cost Smart Energy Monitoring**
Uses affordable sensors (ACS712/SCT-013) and ESP32, delivering smart-meter features at a fraction of commercial smart plug cost.
2. **Real-Time IoT Monitoring**
Shows live voltage, current, power, and energy on a mobile app with graphs, logs, and alerts.
3. **Instant Overload & Abnormal Usage Alerts**
Sends push notifications and can trigger relay cutoff when load crosses safe limits, preventing damage and energy waste.
4. **Remote ON/OFF Control**
Users can control appliances from anywhere using Wi-Fi—turn ON/OFF, schedule, and automate usage.
5. **User-Friendly Dashboard**
App-based interface removes the need for physical meters or analog displays; anyone can monitor usage easily.
6. **Reduces Electricity Bills**
Tracks high-power appliances, shows patterns, and alerts during high consumption, enabling users to optimize usage.
7. **Internet-Connected for Smart Features**
Unlike offline meters, this system fully utilizes IoT—remote monitoring, AI-driven trends, and smart notifications.
8. **Compact, Plug-and-Play Design**
No wiring changes required—just plug it into an outlet and connect your appliance.
9. **Scalable for Multiple Appliances**
More smart plugs can be added to monitor each appliance individually, enabling appliance-level analytics.

DRAWBACKS

Internet Dependency

- Requires Wi-Fi for real-time updates.
- No data access if network is unstable or unavailable.
- Increased delays if internet latency is high.

Accuracy Limitations

- ACS712/SCT sensors have noise and drift; calibration is required.
- ADC accuracy of ESP32 limits precision for very low loads.

CHAPTER – 4

CODING

The coding section of the IoT-based Smart Plug Energy Meter project explains how the ESP32 is programmed to read current values from the ACS712 sensor (and voltage if used), convert them into meaningful electrical parameters, and calculate power consumption using the formula $P = V \times I$. It also describes the initialization of libraries, pin configurations, and the continuous loop that updates readings on the LCD display or sends them to a mobile app/cloud for IoT functionality. The workflow highlights sensor calibration, data conversion, and real-time monitoring, while noting limitations such as noise, load capacity, and network dependency. Finally, the section outlines how the alert system (LEDs, buzzer, or relay control) can be integrated to notify users when power usage exceeds a set threshold, making the code central to the project's functionality.

CODES: -

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// ----> SENSOR PINS <----
#define ZMPT_PIN 34    // Voltage sensor
#define ACS_PIN 35     // Current sensor (ACS712 30A)

// ----> CALIBRATION <----
float voltageScale = 0.048; // Adjusted scale for your voltage sensor
float currentZero = 2048;   // ACS712 midpoint
float currentScale = 0.030; // Adjusted scale for ACS712 30A module

// ----> ENERGY <----
float energy_kWh = 0.0;
unsigned long lastEnergyMillis = 0;

void setup() {
```

```

Serial.begin(115200);
lcd.init();
lcd.backlight();

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Smart Plug Meter");
delay(1000);
}

// ----> GET RMS FUNCTION <----
float getRMS(int pin, float scale, float offset = 0) {
    long sumSq = 0;

    for (int i = 0; i < 500; i++) {
        int raw = analogRead(pin);
        float centered = raw - offset;
        float scaled = centered * scale;
        sumSq += scaled * scaled;
    }

    return sqrt(sumSq / 500.0);
}

void loop() {

    // ----> READ VOLTAGE <----
    float Vrms = getRMS(ZMPT_PIN, voltageScale);

    // ----> READ CURRENT <----
    float Irms = getRMS(ACS_PIN, currentScale, currentZero);

    // ----> REMOVE NOISE WHEN AC IS OFF <----
    if (Vrms < 5) Vrms = 0;
    if (Irms < 0.05) Irms = 0;

    // ----> CALCULATE POWER <----
    float powerW = Vrms * Irms;

    // ----> UPDATE ENERGY <----

```

```

if (millis() - lastEnergyMillis >= 1000) {
    lastEnergyMillis = millis();
    energy_kWh += (powerW / 1000.0) / 3600.0; // Convert W to kWh per second
}

// ----> DISPLAY ON LCD <----
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("V:");
lcd.print(Vrms, 1);
lcd.print(" I:");
lcd.print(Irms, 2);

lcd.setCursor(0, 1);
lcd.print("P:");
lcd.print(powerW, 1);
lcd.print("W E:");
lcd.print(energy_kWh, 3);

delay(600);
}

```

CHAPTER-5

HARDWARE IMPLEMENTATION

1. Microcontroller Board

- **ESP32 or ESP8266 (for IoT and Wi-Fi capability)**
- **Arduino Uno / Arduino Nano (optional for non-IoT version)**

2. Voltage Measurement Components

- **ZMPT101B voltage sensor module (for AC voltage measurement)**
- **High-value resistors for optional voltage divider (if needed)**

3. Current Measurement Components

- **ACS712 current sensor (5A / 20A / 30A depending on load)**
- **Optional: CT Sensor (SCT-013) with burden resistor**

4. Display and Interface Components

- **16×2 LCD Display with I2C module (for real-time readings)**
- **Optional: OLED display for compact alternative**

5. IoT and Connectivity Components

- **Wi-Fi enabled microcontroller (ESP32/ESP8266)**
- **Optional cloud service (Blynk, ThingSpeak) for data logging and remote monitoring**

6. Alert and Indicator Components

- **LED indicators (Red/Yellow/Green) for power usage alerts**
- **Optional: LED bar graph for visual load level (analog)**

7. Miscellaneous Components

- **Resistors (220Ω, 1kΩ) for LED/voltage circuits**
- **Potentiometer (10kΩ) for LCD contrast adjustment**
- **Push buttons (optional) for threshold or relay control settings**

• ACS712 Current Sensor

The **ACS712** is a widely used Hall-effect–based current sensor designed to measure both **AC and DC currents** with electrical isolation. It is commonly used in renewable energy systems, power monitoring circuits, and battery-charging setups where accurate current measurement is essential. The sensor provides a safe interface between high-current loads and low-voltage control systems such as Arduino, ESP32, or other microcontrollers.

The ACS712 module contains a precision Hall-effect IC that detects the magnetic field generated by the current flowing through a built-in copper conductor. This detection is converted into a proportional analog voltage output, which can be easily read by an analog input pin on a microcontroller.

ACS712 sensors come in different versions based on the measuring range, such as **5A, 20A, and 30A**, making them suitable for small, medium, and high-current applications.

Benefits of Using ACS712

- **Electrical Isolation:** The Hall-effect principle ensures isolation between the load and the microcontroller, improving safety.
- **Bidirectional Measurement:** Capable of sensing both positive and negative currents (AC and DC).
- **Analog Output:** Provides a smooth, real-time voltage signal proportional to current flow.
- **Multiple Range Options:** Available in 5A, 20A, and 30A models for varied applications.
- **Compact and Easy to Interface:** Simple wiring and compatibility with most microcontrollers.

Working Concept Overview

1. Current Sensing:

The load current passes through the built-in copper conductor of the ACS712. As current flows, a magnetic field is created around the conductor.

2. Hall-Effect Detection:

A Hall-effect sensor inside the module detects this magnetic field and converts it into a small voltage signal.

3. Output Signal Generation:

The internal circuitry amplifies this signal and outputs an analog voltage proportional to the measured current.

- At **0 A**, the output is approximately $V_{cc}/2$ ($\approx 2.5V$ for a 5V supply).
- Positive current increases the output voltage.
- Negative current decreases it.

4. Microcontroller Processing:

An Arduino or similar controller reads this analog voltage and calculates the actual current using the sensor's sensitivity value (e.g., 185 mV/A for the 5A version).

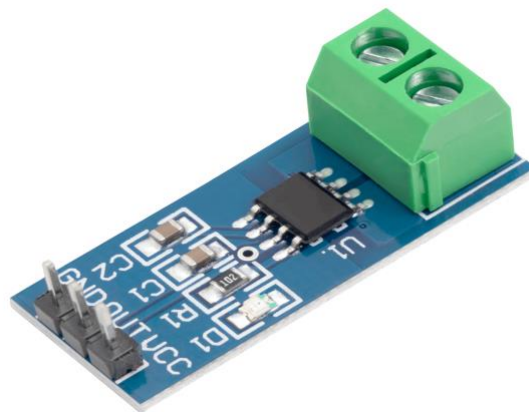


FIG 5.1

ZMPT101B Voltage Sensor Module

The **ZMPT101B voltage sensor module** is a high-precision device designed to measure **AC voltage** safely and accurately. It is widely used in energy monitoring systems, smart home applications, and power analysis projects where precise AC voltage measurement is required. This module provides an isolated low-voltage analog signal that can be easily read by microcontrollers such as Arduino, ESP32, or NodeMCU.

The core of the module is the **ZMPT101B micro transformer**, which allows the AC mains voltage to be stepped down and isolated from the microcontroller. This ensures user safety and protects low-voltage circuitry from high-voltage spikes. The module includes adjustable gain circuitry, allowing precise calibration for different voltage ranges, typically up to **250V AC**.

Benefits of Using ZMPT101B

- **High Accuracy:** Offers excellent measurement precision due to its stable transformer design.
- **Electrical Isolation:** Ensures complete isolation between the AC mains and the microcontroller, improving safety.
- **Adjustable Sensitivity:** The onboard potentiometer allows fine-tuning for accurate readings.
- **Low Noise Output:** Provides a clean analog signal suitable for real-time monitoring.
- **Ideal for AC Monitoring:** Specially designed for measuring household AC voltage (110–250V).

Working Concept Overview

1. AC Voltage Input:

The high AC voltage (such as 110V or 230V) is fed into the primary side of the ZMPT101B transformer.

2. Voltage Transformation:

The transformer steps the voltage down to a much smaller, safe level. This also

ensures **galvanic isolation**, preventing dangerous voltages from reaching the microcontroller.

3. **Signal Conditioning:**

The module contains an operational amplifier and filtering circuitry that convert the small AC signal into a measurable analog output.

- The onboard **potentiometer** allows adjusting the gain to match the expected input voltage.

4. **Microcontroller Interface:**

The conditioned analog signal is read through an analog pin of the Arduino or other controller. Using calibration formulas, the AC voltage amplitude or RMS value can then be calculated.

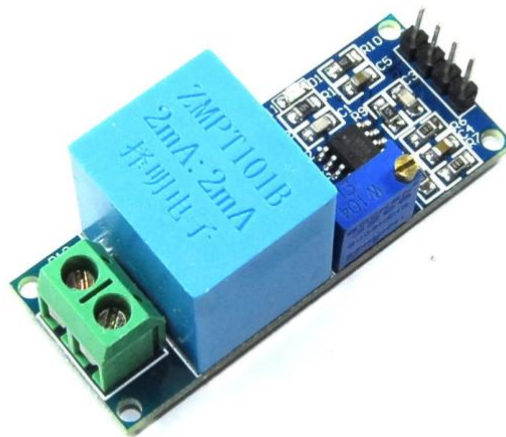


FIG 5.2

ESP32 Microcontroller

The **ESP32** is a powerful and versatile microcontroller widely used in IoT (Internet of Things), automation, robotics, and smart energy systems. Known for its built-in **Wi-Fi and Bluetooth** capabilities, the ESP32 enables seamless wireless communication, remote monitoring, and data transmission, making it ideal for advanced embedded projects.

This dual-core microcontroller integrates a high-performance processor, multiple analog and digital I/O pins, PWM outputs, ADC channels, communication interfaces (UART, I2C, SPI), and low-power operational modes. Its combination of connectivity and processing power makes it suitable for energy monitoring systems, smart solar setups, home automation, and sensor-based applications.

Benefits of Using ESP32

- **Built-In Wi-Fi and Bluetooth:** Enables wireless data transmission, cloud connectivity, and remote control.
- **High Processing Power:** Dual-core Tensilica processor ensures fast and efficient multitasking.
- **Rich Peripheral Support:** ADCs, DACs, PWM, capacitive touch, timers, and communication interfaces.
- **Low-Power Modes:** Ideal for battery-based or energy-efficient systems.
- **Large Community Support:** Extensive libraries and examples simplify development.

Working Concept Overview

1. Data Acquisition:

Sensors such as the ACS712 (current sensor) and ZMPT101B (voltage sensor) send their analog signals to the ESP32's **ADC pins** for measurement.

2. Processing and Analysis:

The ESP32 reads and processes sensor data using internal timers, ADC conversion, and algorithms. It can perform calculations like RMS voltage, current, power, and energy consumption.

3. Wireless Communication:

Using built-in Wi-Fi or Bluetooth, the ESP32 can:

- Send data to cloud platforms or mobile apps

- Host a local web server
- Transmit readings to dashboards for real-time monitoring

4. **System Control:**

The ESP32 can control relays, displays, motors, and other actuators based on sensor feedback or programmed logic.

5. **Low-Power Operation (Optional):**

In energy-sensitive designs, the ESP32 can enter sleep modes to conserve power and wake only for measurements or communication.

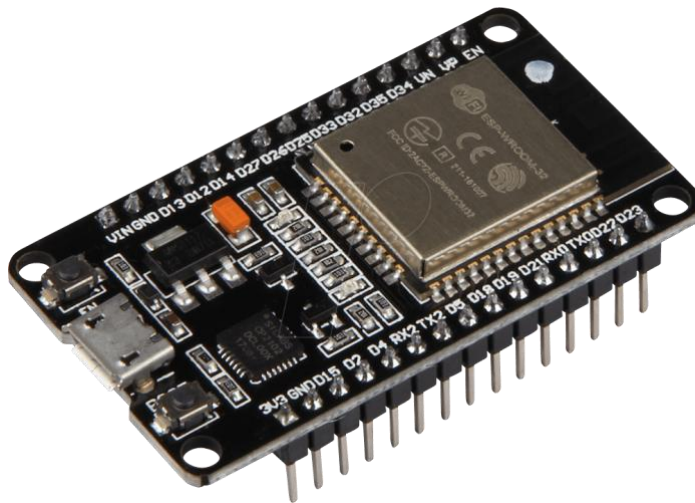


FIG 5.3

Breadboard

A **breadboard** is a fundamental prototyping tool used to build and test electronic circuits without the need for soldering. It provides a reusable platform where components such as resistors, sensors, microcontrollers, and modules can be easily inserted, rearranged, and connected. Breadboards are widely used in educational projects, hobby electronics, and early-stage development of complex systems.

The breadboard consists of interconnected metal clips hidden beneath a grid of holes. These clips create electrical connections between inserted components, allowing circuits to be built quickly and modified as needed. Power rails on the sides supply voltage and ground, while the central terminal strips are used for component connections.

Benefits of Using a Breadboard

- **No Soldering Required:** Components can be easily inserted and removed, making it ideal for experimentation.
- **Reusable and Cost-Effective:** Enables repeated circuit testing without permanent assembly.
- **Easy Prototyping:** Facilitates quick development and troubleshooting of circuits.
- **Structured Layout:** Clearly separated power rails and terminal rows help maintain an organized wiring system.
- **Versatile Compatibility:** Works with microcontrollers like Arduino, ESP32, sensors, LEDs, and various modules.

Working Concept Overview

1. Power Distribution:

The breadboard typically includes two long rows on each side—**power rails**—used to supply **VCC** and **GND** throughout the circuit. These allow easy power access for multiple components.

2. Component Placement:

The central area contains interconnected rows of holes (usually grouped in sets of five). Components such as resistors, jumper wires, ICs, and sensors are placed here to form the circuit connections.

3. Internal Connections:

- The **horizontal rows** in the middle are connected internally, making it easy to connect components together.
- The **vertical rails** on the sides distribute power.

Building the Circuit:

Wires and components are inserted into the holes to form electrical connections. Because everything is temporary, circuits can be easily adjusted, tested, and redesigned.

4. Testing and Debugging:

Once built, the circuit can be powered and tested. Modifications or troubleshooting can be done quickly by rearranging wires or components without soldering.

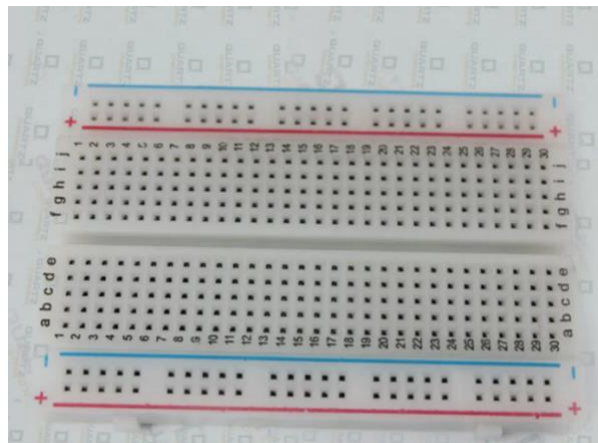


FIG 5.4

BULB HOLDER

A bulb holder is an essential electrical component used to securely mount and connect a light bulb to a power source. It provides both mechanical support and electrical contact, ensuring the bulb operates safely and efficiently. Bulb holders are commonly used in lighting systems, household wiring, testing setups, and project circuits where AC mains bulbs are required. They come in various types such as B22, E27, and E14, depending on the bulb's base. The holder typically includes internal metal contacts that connect to phase and neutral wires, allowing the bulb to receive power when the circuit is activated. In project environments, bulb holders are often used for load testing, energy measurement, and demonstrating power control through relays or sensors.

Benefits of Using a Bulb Holder

- **Safe Electrical Connection:** Ensures that the bulb is properly and securely connected to AC mains.
- **Mechanical Support:** Holds the bulb firmly in place to prevent wobbling or loose connections.
- **Easy Installation:** Simple wiring makes it easy to connect to switches, relays, or monitoring circuits.
- **Compatibility:** Available in various sizes to accommodate different bulb types.
- **Ideal for Testing Loads:** Commonly used in energy monitoring or automation projects as a demonstration load.

Working Concept Overview

1. **Mechanical Mounting:**
The bulb is screwed or locked into the holder, ensuring tight physical contact with the metal terminals.
2. **Electrical Contact:**
Inside the holder, two metal terminals connect to the bulb's base.

- One terminal connects to the phase (live) wire.
- The other terminal connects to the neutral wire.

3. Power Supply:

When power is supplied, current flows through the holder's terminals into the bulb's filament or LED driver circuit, causing it to illuminate.

4. Integration in Projects:

In monitoring systems or automation setups:

- A relay or controller can switch the bulb ON/OFF.
- Sensors (like ACS712 or ZMPT101B) can measure the bulb's current or voltage.
- The holder acts as the safe interface between the AC bulb and the project circuitry.



FIG 5.5

LCD Display

An **LCD (Liquid Crystal Display)** is a widely used output device in electronics projects for displaying text, numbers, sensor readings, and system status information. Commonly found in embedded systems, monitoring devices, and user interfaces, LCD displays provide a clear and energy-efficient method for presenting real-time data. The most popular type used in microcontroller projects is the **16×2 LCD**, which can display 16 characters per line across two lines.

LCDs operate by controlling liquid crystal molecules that either block or allow light to pass through, forming visible characters. These displays often use the **HD44780 controller**, making them easy to interface with microcontrollers such as Arduino and ESP32. Many modules also include an **I2C interface**, reducing the number of required wiring connections from 16 pins to just 4.

Benefits of Using an LCD Display

- **Clear Visual Output:** Displays sensor data, messages, and system parameters in real time.
- **Low Power Consumption:** Consumes minimal energy, making it ideal for portable or solar-powered systems.
- **Wide Compatibility:** Works seamlessly with microcontrollers like ESP32, Arduino Uno, and Raspberry Pi.
- **Optional Backlight:** Allows visibility in low-light conditions.
- **Easy Programming:** Supported by numerous libraries such as *LiquidCrystal* and *LiquidCrystal_I2C*.

Working Concept Overview

1. Character Formation:

Liquid crystals inside the display change their orientation based on electrical signals, creating characters and symbols on the screen.

2. Controller Interface:

Most LCDs use the HD44780 controller, which manages the display memory and character generation.

- **Parallel Mode:** Requires multiple digital pins for data and control.
 - **I2C Mode:** Uses only SDA and SCL pins, simplifying connectivity.
3. **Backlight Operation:**
An LED backlight illuminates the display from behind, improving readability.
4. **Microcontroller Communication:**
The ESP32 or Arduino sends commands and data to the LCD:
- Commands control cursor position, clearing the screen, or changing display modes.
 - Data writes alphanumeric characters for display.
5. **Real-Time Displaying:**
Sensor readings like voltage (from ZMPT101B) or current (from ACS712) can be shown continuously, allowing users to monitor system operation instantly.



FIG 5.6

Circuit Breakdown

Components:

- ESP32 / ESP8266 microcontroller
- ACS712 Current Sensor (5A / 20A / 30A)
- ZMPT101B Voltage Sensor (for AC mains)
- 16×2 LCD Display with I2C module (or OLED display optional)
- LED indicators (Red, Yellow, Green)
- Jumper wires
- Optional: Relay module (for smart plug control)
- Test Load: 40W LED Bulb

Circuit Diagram Overview

- Voltage and Current Connections:
 - Voltage sensor output connected directly to ESP32 analog pin.
 - ACS712 current sensor output connected to another analog pin.
 - 40W LED bulb plugged into smart plug for testing.
- Microcontroller Logic:
 - ESP32 continuously reads voltage and current values.
 - Computes real-time power: $\text{Power} = \text{Voltage} \times \text{Current}$.
 - Integrates power over time to calculate total energy consumption (kWh).
- Display and Alert System:
 - LCD displays Voltage (V), Current (A), Power (W), Energy (kWh) in real time.
 - LED indicators light up when power consumption crosses predefined thresholds.
- IoT/Remote Monitoring (Optional):

- ESP32 sends real-time readings to a mobile app or cloud dashboard.
- Provides remote monitoring and notifications when energy exceeds set limits.
- Operation Summary with 40W LED Bulb:
- The LED bulb is plugged into the smart plug.
- Current sensor measures current drawn by the bulb ($\approx 0.18\text{A}$ at 230V AC).
- Voltage sensor measures mains voltage.
- ESP32 calculates instantaneous power ($\sim 41\text{W}$) and updates cumulative energy consumption (kWh).
- LED indicators or LCD display alert the user if energy usage exceeds thresholds.
- Optional relay can automatically cut off the bulb if power exceeds safety limits.
- Explanation of How It Works (Logic Flow):
- ESP32 reads analog inputs from ACS712 and ZMPT101B.
- Analog values are converted to RMS voltage and current.
- Instantaneous power is calculated: $P = V \times I$.
- Energy is integrated over time to get kWh.
- LCD updates readings continuously, and LEDs provide visual alerts.
- Optional: IoT module sends data to cloud/mobile for remote monitoring and analytics.

CIRCUIT DIAGRAM

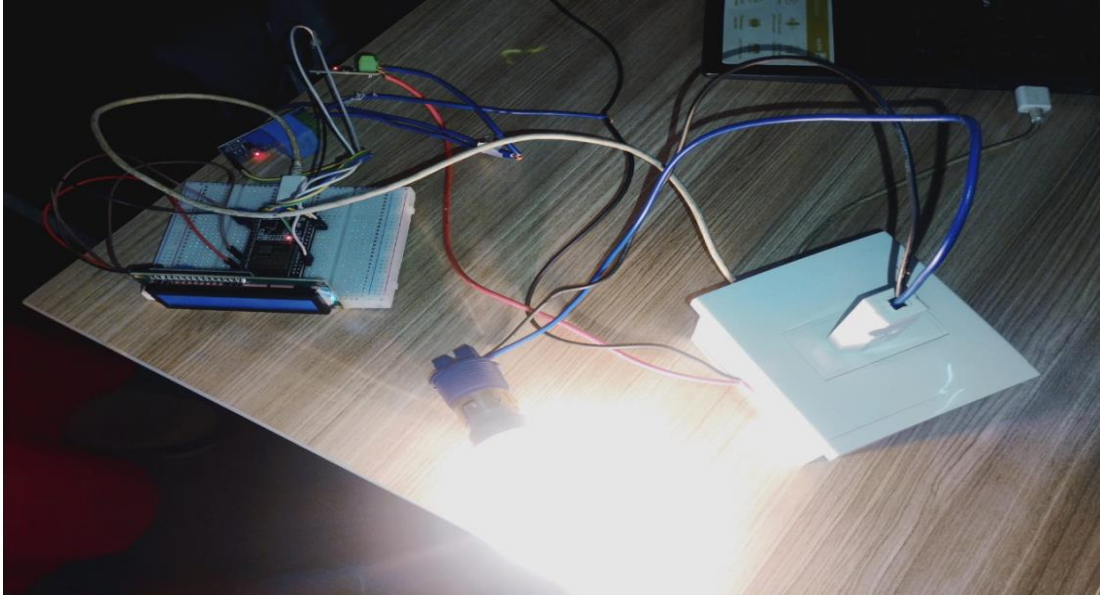


FIG 5.8

CHAPTER-6

WORKING

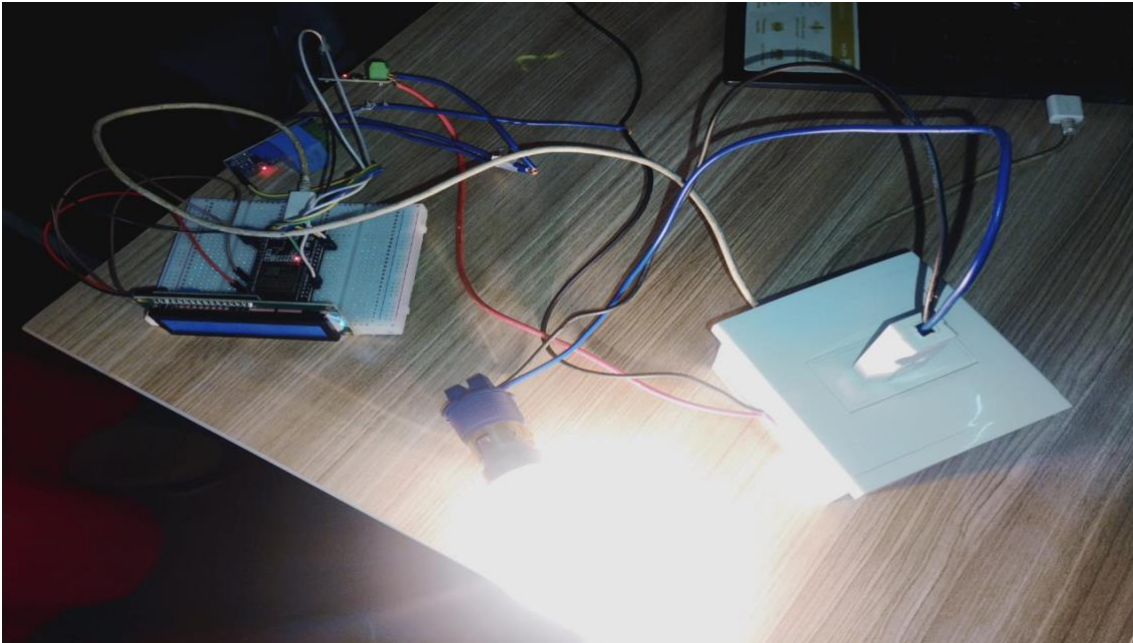


FIG 6.1

The working of the IoT-Based Smart Plug Energy Meter begins when a 230V AC supply powers the connected load, such as a 40W LED bulb. As soon as the bulb is switched ON, current starts flowing through it. A current sensor (ACS712) is connected in series with the load to measure this current. The sensor continuously detects the current drawn by the bulb and converts it into a proportional analog voltage signal. This signal is sent to the ESP32 microcontroller through its ADC (Analog-to-Digital Converter) pin.

The microcontroller processes the data to calculate real-time electrical parameters. Using the standard formula, instantaneous power is computed as:

$$\text{Power (W)} = \text{Voltage (V)} \times \text{Current (A)}$$

The total energy consumed over time is calculated as:

$$\text{Energy (kWh)} = \text{Power} \times \text{Time}$$

These values are displayed on the LCD in real time and can optionally be sent to a mobile app or cloud platform for IoT monitoring. LED indicators alert the user if power consumption exceeds predefined thresholds, while an optional relay can automatically disconnect the load to prevent overconsumption. This setup enables accurate, real-time monitoring, control, and energy management for household appliances.

TEAM MEMBERS'



RESULT

The IoT-Based Smart Plug Energy Meter was successfully designed and implemented using an ESP32 microcontroller, ACS712 current sensor, ZMPT101B voltage sensor, and a 16×2 LCD display. The system accurately measured real-time voltage, current, power, and cumulative energy consumption of a household load, demonstrated here using a 40W LED bulb.

The LED indicators and LCD display effectively notified the user when power usage exceeded predefined thresholds, enabling immediate corrective actions such as switching off the connected appliance. Testing showed that the system:

- Provides real-time monitoring of voltage, current, power, and energy usage
- Triggers instant alerts when energy consumption exceeds limits
- Delivers stable and consistent readings using low-cost sensors
- Offers a user-friendly and affordable solution suitable for homes, hostels, and small communities
- Helps in reducing unnecessary electricity consumption, as verified during practical testing

Overall, the developed smart plug energy meter meets the objectives of being affordable, simple, and practical for daily household energy management, serving as a cost-effective alternative to commercial smart plugs or meters.

CONCLUSION

The **IoT-Based Smart Plug Energy Meter** provides an efficient, low-cost method for monitoring and controlling household electricity consumption. Using the ACS712 current sensor and ZMPT101B voltage sensor, the system calculates real-time power and energy usage, displaying the data on an LCD. LED indicators alert the user whenever the connected appliance exceeds preset energy thresholds.

The project achieves its main objectives by:

- Allowing users to track energy consumption of individual appliances
- Helping prevent wastage and reduce electricity bills
- Offering a simple and user-friendly interface without requiring complex programming or internet access

This system is highly suitable for households, hostels, and small communities, particularly where cost-effective and easy-to-use energy monitoring is needed. It also lays the foundation for future enhancements such as IoT-based remote monitoring, data logging, and smart automation.

FUTURE SCOPE

Future versions can incorporate Wi-Fi or GSM modules to enable remote monitoring through mobile apps or cloud dashboards, allowing users to track consumption from anywhere. The system can be extended to monitor individual household appliances, helping users identify which devices consume the most energy. By adding memory storage or SD card modules, long-term consumption data can be recorded, enabling energy usage analysis and prediction of monthly electricity bill. The system can be upgraded to automatically turn off appliances or send alerts via SMS/app when abnormal energy usage is detected.

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